

Question 4: Inverter-Chain Delay Optimization with Non-Ideal Constraints [20 points]

A minimum-size inverter has input capacitance

$$C_{in,min} = 10fF.$$

You must drive a load

$$C_L = 6.4pF.$$

Use the normalized inverter delay model

$$d_i = f_i + \gamma$$

where

$$\gamma = 1.$$

For an inverter chain, the effective fanout of stage i is

$$f_i = \frac{C_{in,i+1}}{C_{in,i}}$$

for intermediate stages, and

$$f_N = \frac{C_L}{C_{in,N}}$$

for the final stage.

(a) **Electrical effort.** Compute the total electrical effort

$$F = \frac{C_L}{C_{in,min}}.$$

↑
fanout

[3 points]

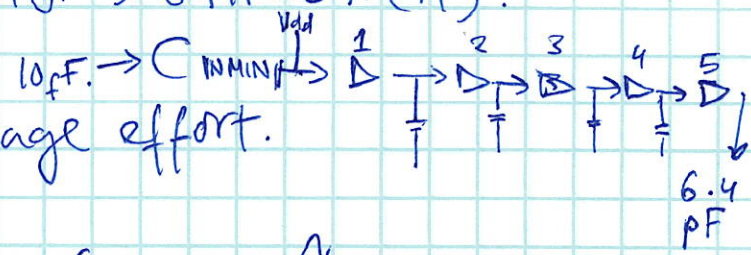
$$\frac{6.4pF}{10fF} = \frac{6400fF}{10fF} = \underline{640}$$

Q4c

$N=5$. equal stage effort $\approx 3.6411284 \cdot (A)$.

$$f_N = f_5 = \frac{C_L}{C_{IN,5}}$$

$f = \text{stage effort}$.



$$A = \frac{6.4 \text{ pF}}{C_5} \rightarrow C_5 = \frac{6.4 \text{ pF}}{A} \approx 1.7576969 \text{ pF}$$

$$C_4 = \frac{C_{IN,5}}{A} \approx 0.4827341 \text{ pF}$$

$$C_3 = \frac{C_{IN,4}}{A} = 0.1325782 \text{ pF}$$

$$C_2 = \frac{C_{IN,3}}{A} = 0.0364113 \text{ pF}$$

$$C_{IN,1} = \frac{C_{IN,2}}{A} = 0.0100000 \text{ pF} = C_{IN,1}$$

(hopefully equals $C_{in,min}$)

$$C_{IN,MIN} = C_{IN,1} = 10 \text{ fF}$$

$$1 \text{ pF} = 1000 \text{ fF}$$



Notes Pg 11, Q4c

- (b) **Ideal number of stages.** Using the rule of thumb that a good stage effort is near 4, estimate the best number of inverter stages. Justify your answer. [4 points]

5 gates/stages -

stage effort per inverter closer to 4 when 5 gates than 4 gates

Working on Notes Page 10
Reasoning on Notes Page 10

- (c) **Continuous sizing.** For your chosen number of stages N , compute the equal stage effort

$$f = F^{1/N}.$$

Then compute the input capacitance of each inverter stage in the chain, starting with the minimum-size inverter. [5 points]

$$N = 5. \quad F^{1/5} = f = 3.6411284$$

$\nearrow 640^{1/5}$ (see pg 10, Q4B for working for this)

see notes page 11
For working please!
 $\uparrow$
\$ answers

4d)

Notes Pg 12

~~$$F_1 = \frac{C_{IN1}}{C_{IN MIN}} = \frac{10 \text{ fF}}{10 \text{ fF}} = 1$$~~

~~F_2~~ Start from end!

Effective fanout

$$F_5 = \frac{6400 \text{ fF}}{C_{IN5}} = \frac{6400}{1280} = 5 \text{ electrical fanout.}$$

$$F_4 = \frac{C_{IN5}}{C_{IN4}} = \frac{1280}{320} = 4 \text{ electrical fanout.}$$

$$F_3 = \frac{C_{IN4}}{C_{IN3}} = \frac{320}{80} = 4 \text{ electrical fanout.}$$

$$F_2 = \frac{C_{IN3}}{C_{IN2}} = \frac{80}{20} = 4 \text{ electrical fanout.}$$

$$F_1 = \frac{C_{IN2}}{C_{IN1}} = \frac{20}{10} = 2 \text{ electrical fanout.}$$

$$d_i = f_i + 1, \quad 1 \leq i \leq 5.$$

$$\left. \begin{aligned} d_1 &= 2+1=3 \\ d_2 &= 4+1=5 \\ d_3 &= 4+1=5 \\ d_4 &= 4+1=5 \\ d_5 &= 5+1=6 \end{aligned} \right\}$$

sums to: 24 tau

↑
unit of time?

ask yang Wu

$$15+9=24$$

total normalized delay